

DATA CLEANING & PREPARATION REPORT

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Project: Project 1 – Data Cleaning & Preparation

Tool Used: Python (Pandas)

Objective

The objective of this project was to clean and prepare a raw dataset by identifying missing values, checking for duplicate records, validating data formats, and creating a clean dataset suitable for analysis.

Dataset Overview

- Total Records: 1200
- Total Columns: 14
- File Format: Excel (.xlsx)

Data Quality Assessment

1. Missing Values Analysis

The dataset was examined for missing values using Pandas.

Findings:

- CouponCode column contained 309 missing values.
- All other columns had no missing values.

2. Missing Value Treatment

The missing values in the CouponCode column were handled by replacing them with the value:

"No Coupon"

This approach preserved all records while maintaining data consistency.

3. Duplicate Record Analysis

The dataset was checked for duplicate records.

Findings:

- Duplicate Rows Found: 0

- Duplicate Order IDs Found: 0

No duplicate records were present in the dataset.

4. Data Format Verification

The dataset was inspected to ensure consistency in:

- Date formats
- Numeric values
- Text fields

Findings:

- Date formats were valid and consistent.
- Numeric fields were correctly formatted.
- Text fields were properly structured.

No format corrections were required.

Cleaning Actions Performed

Action	Status
Identified Missing Values	Completed
Filled Missing Coupon Codes	Completed
Checked Duplicate Rows	Completed
Verified Unique IDs	Completed
Verified Date Formats	Completed
Saved Cleaned Dataset	Completed

Final Verification

Metric	Result
Remaining Missing Values	0
Duplicate Rows	0
Duplicate Order IDs	0
Incorrect Date Formats	0

Output

A cleaned dataset was generated and saved as:

Cleaned_Dataset.xlsx

Conclusion

The dataset was successfully cleaned and prepared for further analysis. Missing values were handled appropriately, duplicate records were verified, data formats were validated, and a clean dataset was produced for future analytical tasks.